ICDFA | WADF - 2026-M01 | Linux Essentials | Week 03

LAB 5 & LAB 6 EVIDENCE REPORT

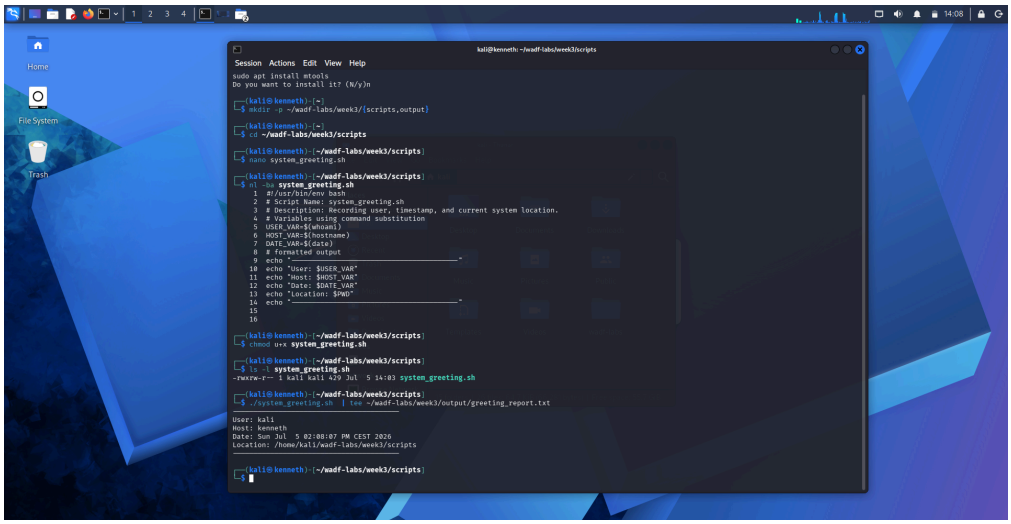
10 July 2026, 23:59 WAT

Student identification

Full name	KENNETH UKATA
Registration number	fcdf2617034
Linux distribution / version	DEBIAN LINUX
VM hostname	kali@KENNETH

LAB 5A:

```
#!/usr/bin/env bash
# Script Name: system_greeting.sh
# Description: Recording user, timestamp, and current system location.
# Variables using command substitution
USER_VAR=$(whoami)
HOST_VAR=$(hostname)
DATE_VAR=$(date)
# formatted output
echo "----------------------------------------"
echo "User: $USER_VAR"
echo "Host: $HOST_VAR"
echo "Date: $DATE_VAR"
echo "Location: $PWD"
echo "----------------------------------------"
```



LAB 5B:

```
#!/usr/bin/env bash

# Script Name: create_notes.sh

# Description: Validates input and creates three numbered note files.

# Accept project name as a positional parameter

project_name="$1"

# Validation control: Check if blank

if [ -z "$project_name" ]; then

echo "Error: Project name cannot be blank." >&2

exit 1

fi

# Validation control; Check for unsafe path separators

if [[ "$project_name" == *"/"* || "$project_name" == *".."* ]]; then

echo "Error: Unsafe path separators detected." >&2

exit 1

fi

# Target directory setup

TARGET_DIR="$HOME/wadf-labs/week3/genarated-notes/$project_name"
```

```
mkdir -p "$TARGAT_DIR"

# For loop to generate note files

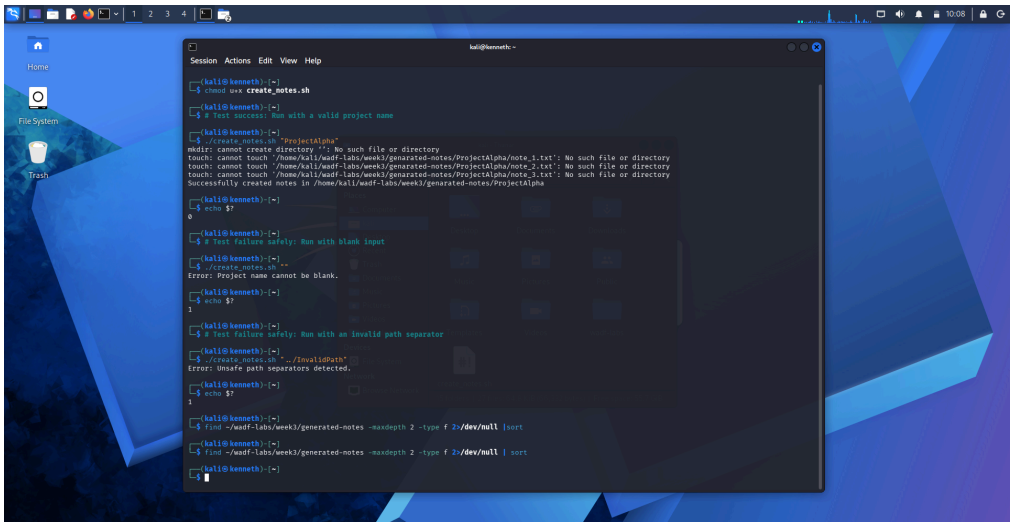
for i in {1..3}; do

touch "$TARGET_DIR/note_${i}.txt"

done

echo "Successfully created notes in $TARGET_DIR"

exit 0
```



LAB 5C:

SYSTEM INVENTORY REPORT

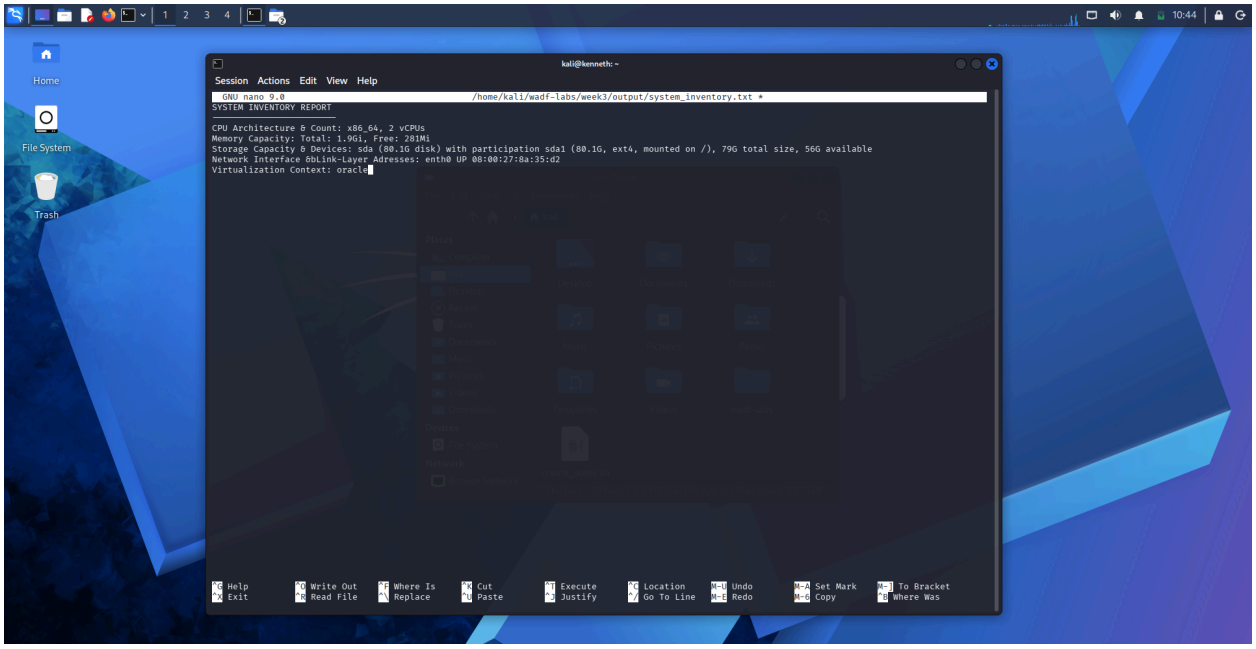
CPU Architecture & Count: x86_64, 2 vCPUs

Memory Capacity: Total: 1.9GiB, Free: 281Mi

Storage Capacity & Devices: sda (80.1G disk) with partition sda1 (80.1G, ext4, mounted on /), 79G total size, 56G available

Network Interface & Link-Layer Addresses: eth0 UP 08:00:27:8a:35:d2

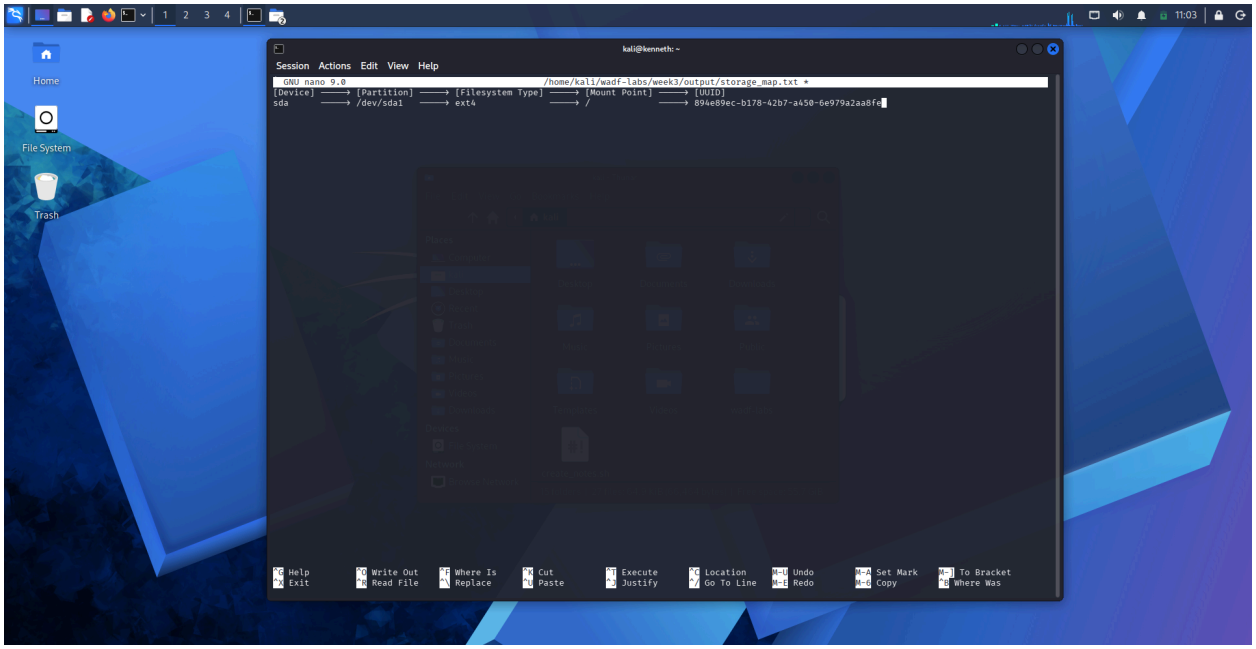
Virtualization Context: oracle



LAB 6A:

[Device] -----> [Partition] -----> [Filesystem Type] -----> [Mount Point] -----> [UUID]

sda -----> /dev/sda1 -----> ext4 -----> / -----> 894e89ec-b178-42b7-a450-6e979a2aa8fe



LAB 6B:

DATA LOCATION MAP WORKSHEET

User files: /home

System configuration: /etc

Logs: /var/log

Inatalled software: /usr (or /opt)

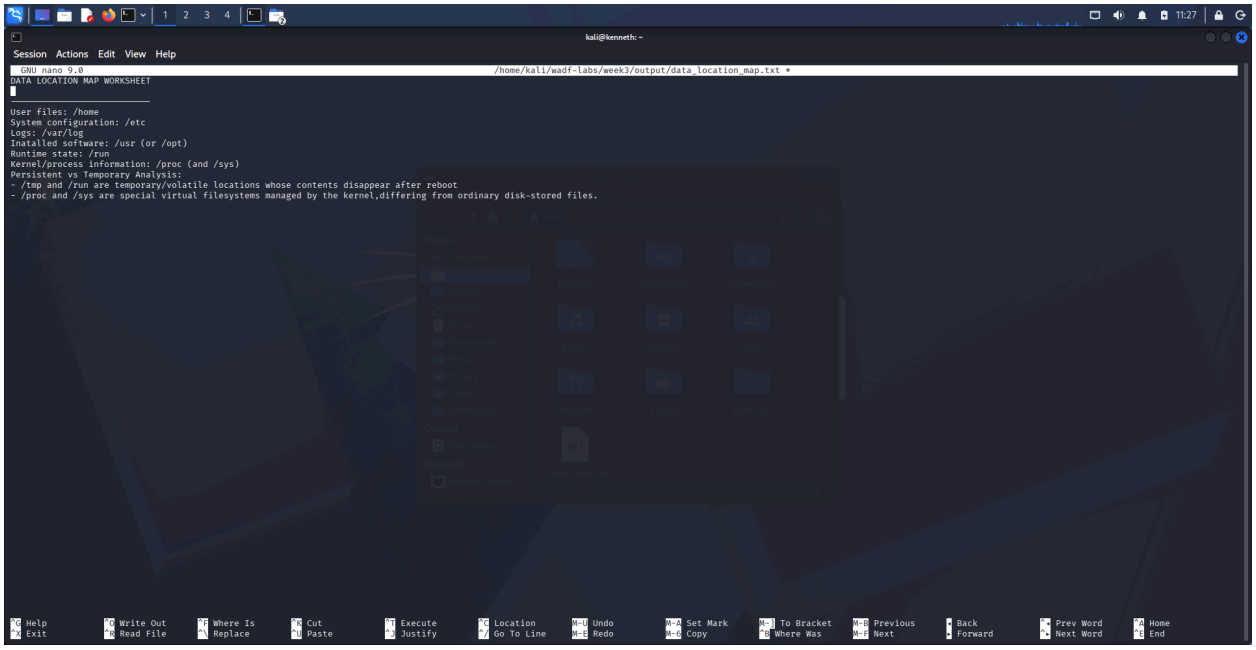
Runtime state: /run

Kernel/process information: /proc (and /sys)

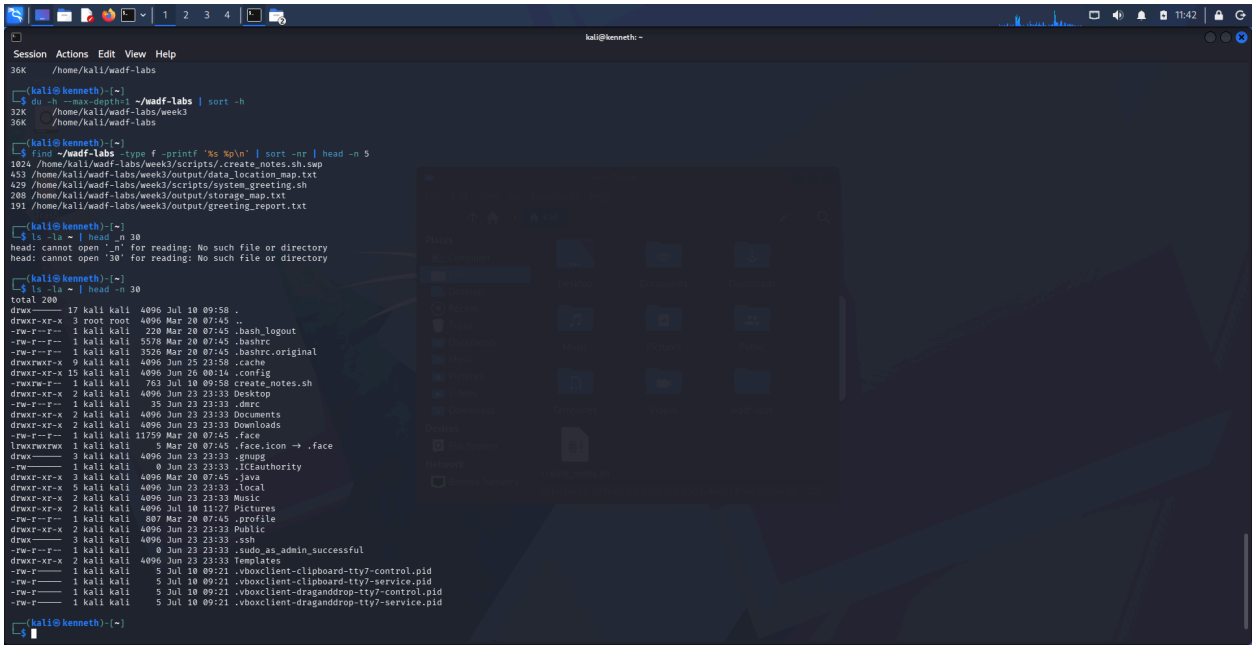
Persistent vs Temporary Analysis:

- /tmp and /run are temporary/volatile locations whose contents disappear after reboot

- /proc and /sys are special virtual filesystems managed by the kernel,differing from ordinary disk-stored files.



LAB 6C:



Summary:
In this week lab we focused on automating administration tasks and inspecting Linux system architecture using safe, read-only commands. In Lab 5, I successfully built two Bash scripts featuring command substitution, variables, and input validation to check for blank names or unsafe paths before creating files. In Lab 6, I analyzed where Linux stores data, verified that my system runs inside an Oracle virtual machine, and mapped my primary storage device () using its exact UUID. it was confusing but exciting exercise .